

Hierarchical Bayesian modeling of multi-region brain cell count data

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This study proposes an **important** new approach to analyzing cell-count data, which are often undersampled and cannot be accurately assessed using traditional statistical methods. The case studies presented in the article provide **compelling** evidence of the superiority of the proposed methodology over existing approaches, which could promote the use of Bayesian statistics among neuroscientists. The authors have taken steps to make the methodology accessible, although some implementation difficulties are likely to remain.

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Abstract

We can now collect cell-count data across whole animal brains quantifying recent neuronal activity, gene expression, or anatomical connectivity. This is a powerful approach since it is a multi-region measurement, but because the imaging is done post-mortem, each animal only provides one set of counts. Experiments are expensive and since cells are counted by imaging and aligning a large number of brain sections, they are time-intensive. The resulting datasets tend to be under-sampled with fewer animals than brain regions. As a consequence, these data are a challenge for traditional statistical approaches. We present a 'standard' partially-pooled Bayesian model for multi-region cell-count data and apply it to two example datasets. These examples demonstrate that hierarchical Bayesian methods are well suited to these data. In both cases the Bayesian model outperformed standard parallel *t*-tests. Overall, inference for cell-count data is substantially improved by the ability of the Bayesian approach to capture nested data and by its rigorous handling of uncertainty in under-sampled data.

Introduction

In studying the brain we are often confronted with phenomena that involve specific subsets of neurons distributed across many brain regions. Computations, for example, are performed by neuronal networks connecting cells in different parts of the brain. As another example, from development, neurons in different anatomical regions of the brain share the same lineage. Data for each of these types of experiment will be considered here, but the challenge is very general: how to measure and analyze multi-region neuronal data with cellular resolution.

In a typical cell-count experiment, gene expression is used to tag the specific cells of interest with a targeted indicator [Kawashima et al. \(2014\)](#). The brain is sliced, an entire stack of brain sections from a single animal is imaged and the images are aligned and registered to a standardized brain atlas such as the Allen mouse atlas ([Lein et al., 2007](#); [Oh et al., 2014](#); [Daigle et al., 2018](#); [Harris et al., 2019](#)), the images are segmented into anatomical regions and the labeled cells in each region are counted. The resulting dataset consists of labeled cell counts across each of ~10–100 brain regions. This technology is being deployed to address questions in a broad range of neuroscience subfields, for example: memory ([Kim and Cho, 2017](#); [Haubrich and Nader, 2023](#); [Dorst et al., 2024](#)), neurodegenerative disorders ([Liebmann et al., 2016](#)), social behavior [Kim et al. \(2015\)](#), and stress ([Bonapersona et al., 2022](#)).

Cell-counts are often compared across groups of animals which differ by an experimental condition such as drug treatment, genotype or behavioural manipulation. However, the expense and difficulty of the experiment means that the number of animals in each group is often small. Ten is a typical number of sample for these experiments, but fewer is not uncommon. This means that these data are undersampled: the dimensionality of the data, which corresponds to the number of brain regions, is much larger than the number of samples, which usually corresponds to the number of animals (**Figure 1A**). Current statistical methods are not well-suited to these nested wide-but-shallow datasets. Furthermore, because of the complicated preparation and imaging procedure, there is often missing data along with variability derived from experimental artifacts.

In cell count data there are two obvious sources of noise. The first of these is easy to describe: if a region has a rate that determines how likely a cell is to be marked for counting, then the actual number of marked cells is sampled from a Poisson distribution. The second source of noise is the animal to animal variability of the rate itself and this depends on diverse features of the individual animal and the experiment that are often unrelated to the phenomenon of interest. The challenge is to control for outliers and ‘poor’ data points whose rate is noisy, while extracting as much information as possible about the underlying process. Dealing with outliers is often an opaque and *ad hoc* procedure. It is also a binary decision, a point is either excluded so it does not contribute, or included, noise and all. This is where partial pooling helps. Partial pooling allows for the simultaneous estimation of parameters describing individual data points and parameters describing populations. This helps the data to self-regularize and elegantly balances the contribution of informative and weak observations to parameter values.

In recent years Bayesian approaches to data analysis have become powerful alternatives to classical frequentist approaches ([Gelman et al., 2013](#); [McElreath, 2018](#); [van de Schoot et al., 2021](#)). They have been applied to some types of neuroscience data including neurolinguistics ([Dimmock et al., 2023](#)), neural coding ([Brown et al., 1998](#); [Zhang et al., 1998](#)), synaptic parameters ([Moran et al., 2008](#); [Costa et al., 2013](#); [Bird et al., 2016](#); [Bykowska et al., 2019](#)) and neuronal-circuit connectivity ([Mishchencko et al., 2011](#); [Cinotti and Humphries, 2022](#)). A Bayesian approach is particularly well suited to cell-count data but has not previously been applied to this problem.

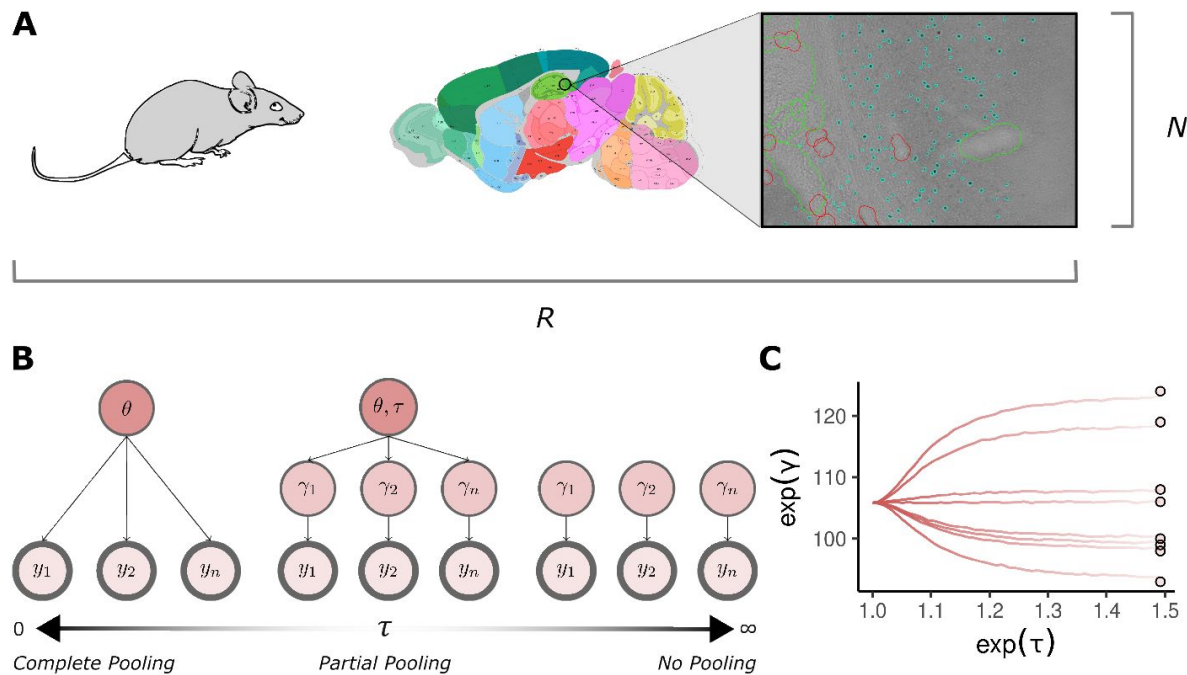


Figure 1.

Introduction.

A: Each of N animals produce a cell-count from a total of R brain regions of interest. Cell-count data is typically undersampled with $N \ll R$. Scientists analyse the brain sections from experiment for positive signals. Here an example section is shown where teal points mark cells expressing the immediate early gene *c-Fos* (green and red lines indicate regions labeled as damaged). The final cell-count is equal to the sum of these individual items (sagittal brain map taken from the Allen mouse brain atlas: <https://mouse.brain-map.org>). **B:** Partial pooling is a hierarchical structure that jointly models observations from some shared population distribution. It is a continuum that depends on the value of the population variance τ . When $\tau = 0$ there is no variation in the population and each individual observation is modeled as a conditionally independent estimate of some fixed population mean θ (complete pooling). As τ tends to infinity observations do not combine inferential strength but inform an independent estimate y_i (no pooling). In between the two extremes combine. Each observation can contribute to the population estimate while simultaneously supporting a local one to effectively model the variance in the data. The observed data quantities, y_1 to y_n , are highlighted with a thick line in the model diagrams. **C:** An example of partial pooling on simulated count data. As the population standard deviation increases on the x-axis the individual estimates $\exp(y_i)$ trace a path from a completely pooled estimate to an unpooled estimate. Circular points give the raw data values. Parameters are exponentiated because the outcomes are Poisson and so parameters are fit on the log scale.

A Bayesian approach formalizes the process of scientific inference, it distinguishes the data and a probabilistic mathematical model of the data. This model has a likelihood which gives the probability of the observed data for a given set of model parameters. The model often has a hierarchical structure which we compose to reflect the structure of the experiment and the investigators hypothesis of how the data depends on experimental condition. This hierarchy determines a set of *a priori* probabilities for the parameter values. The result of Bayesian inference is a probability distribution for these model parameters given the data, termed the posterior.

There are three advantages of a Bayesian approach that we want to emphasize, (1) while traditional multi-level models also allow a hierarchy (Aarts et al., 2014 [↗](#)), Bayesian models are more flexible and the role of the model is clearer, (2) since the result of Bayesian inference is a probability distribution over model parameters, it indicates not just the fitted value of a parameter but the uncertainty of the parameter value. Finally (3) Bayesian models tend to make more efficient use of data and therefore improve statistical power.

A Bayesian model also includes a set of probability distributions, referred to as the prior, which represent those beliefs it is reasonable to hold about the statistical model parameters before actually doing the experiment. The prior can be thought of as an advantage, it allows us to include in our analysis our understanding of the data based on previous experiments. The prior also makes explicit in a Bayesian model assumptions that are often implicit in other approaches. However, having to design priors is often considered a challenge and here we hope to make this more straight-forward by suggesting priors that are suitable for this class of data.

Here, our aim is to introduce a ‘standard’ Bayesian model for cell count data. We illustrate the application of this model to two datasets, one related to neural activation and the other to developmental lineage. For the second dataset, we also demonstrate a second example extension Bayesian model. In all cases, the Bayesian models produce clearer results than the classical frequentist approach.

Materials and Methods

Data

To illustrate our approach we consider two example applications, one which counts cells active in regions of the recognition memory circuit of rat during a familiarity discrimination task, and the other which examines the distribution of a specific interneuron type in the mouse thalamus.

Case study one - transient neural activity in the recognition memory circuit

The recognition memory network, **Figure 3** [↗](#), is a distributed network which has been well studied using a variety of behavioural tasks. It includes the hippocampus (HPC) and perirhinal cortex (PRH), shown to deal with object spatial recognition and familiarity discrimination respectively (Barker and Warburton, 2011 [↗](#); Ennaceur et al., 1996 [↗](#); Norman and Eacott, 2004 [↗](#)); medial prefrontal cortex (MPFC), concerned with executive functions such as decision making but also with working memory, and the temporal association cortex (TE2) used for acquisition and retrieval of long-term object recognition memories (Ho et al., 2011 [↗](#)). The nucleus reuniens (NRe) has reciprocal connectivity to both MPFC and HPC (Hoover and Vertes, 2012 [↗](#)) and for this reason it is also believed to be an important component of the circuit (Barker and Warburton, 2018 [↗](#), 2011 [↗](#)). In previous studies, lesions of the NRe have been shown to significantly impair long-term but not short-term object-in-place recognition memory (Barker and Warburton, 2018 [↗](#)).

The data analysed in this case study were collected to investigate the role of the NRe in the recognition memory circuit through contrasting the neural activation for animals with a lesion in the NRe with neural activation for animals with a sham surgery. The immediate early gene *c-Fos* is rapidly expressed following strong neural activation and is useful as a marker of transient neural activity. Animals in the experiment performed a familiarity discrimination task (single-item recognition memory), discriminating novel or familiar objects with or without an NRe lesion and the number of cells that expressed *c-Fos* was counted in regions across the recognition memory circuit. The two-by-two experimental design allocated ten animals to each of the four experiment groups {sham, lesion} × {novel, familiar} and cell counts were recorded from a total of 23 brain regions. The visual cortex V2C and the motor cortex M2C were taken as control regions as they were not expected to show differential *c-Fos* expression in response to novel or familiar objects (Exley, 2019).

Case study two - Ontogeny of inhibitory neurons in mouse thalamus and hypothalamus

The second dataset comes from a study Jager et al. (2021) that, in part, counted the number of inhibitory interneurons in the thalamocortical regions of mouse. *Sox14* is a gene associated with inhibitory neurons in subcortical areas. It is required for the development and migration of local inhibitory interneurons in the dorsal lateral geniculate nucleus (LGd) of the thalamus Jager et al. (2016, 2021). Consequently, *Sox14* is useful for identifying discrete neuronal populations in the thalamus and hypothalamus (Golding et al., 2014; Jager et al., 2016, 2021). The experiment compared heterozygous (HET) and knock-out (KO) mouse lines. The HET knock-in mouse line *Sox14^{GFP/+}* marked *Sox14* expressing neurons with green fluorescent protein (GFP), the homozygous KO mouse line *Sox14^{GFP/GFP}*, in contrast, was engineered to block expression of the endogenous *Sox14* coding sequence Delogu et al. (2012). Each animal produced two samples, one for each hemisphere. In total there are ten data points, six belonging to HET (three animals), and four to KO (two animals). Each observation is 50 dimensional, corresponding to fifty individual brain regions in each hemisphere.

Hierarchical modeling

Our goal in both cases is to quantify group differences in the data. We present a ‘standard’ hierarchical model. This model reflects the experimental features common to cell count experiments and reflects the hierarchical structure of cell count data; the standard model is designed to deal robustly and efficiently with noise. On some occasions, to reflect a specific hypotheses, the structure of a particular experiment or an observed source of noise, this model can be further refined or changed to target the analysis. We will give an example of this for our second dataset.

At the bottom of the model are the data themselves, the cell counts y_i . The index i runs over the full set of samples, which in this case comprises 23 brain regions × 10 animals × 4 groups ≈ 920 datapoints in the first study, and 50 brain regions × 6 HET animals + 50 brain regions × 4 KO animals ≈ 500 datapoints in the second. The basic assumption the model makes is that this count is derived from an underlying propensity, $\lambda_i > 0$, which depends on brain region and, potentially, group:

$$y_i \sim \text{Poisson}(\lambda_i) \quad (1)$$

Hence the propensity λ_i is the mean of the Poisson distribution, and a statistical model is used to describe the dependence of this parameter on brain region and animal. Since λ_i is strictly positive a log-link function is introduced:

$$\log \lambda_i = \theta_{r[i],g[i]} + \gamma_i + E_i \quad (2)$$

where we have used ‘array notation’ [Gelman and Hill \(2006\)](#), mapping the sample index i to properties of the sample, so $r[i]$ returns the region index of observation i , and similar for $g[i]$ but for groups and animals. The sample-by-sample variability is given by γ_i , this is modelled as Gaussian noise:

$$\gamma_i \sim \text{Normal}(0, \tau_{r[i],g[i]}) \quad (3)$$

whose size depends on region and group. This equation demonstrates a potentially surprising aspect of partially-pooled models: the over-parameterization.

Ignoring E_i for now, the rate has been split between two terms, $\theta_{r[i],g[i]}$ is the *fixed effect* which is constant across animals and $\gamma_{r[i],g[i]}$ is the *random effect* which captures the animal to animal variability. While $\theta_{r,g}$ models the mean for log-cell-count for each region, given the condition; γ_i models variation around this mean. For this reason, γ_i is assumed to follow a normal distribution with zero mean. The regression term may appear over-parameterized, without $\theta_{r,g}$ the γ_i could ‘do the work’ of matching the data. However, the model is regularized by a prior; observations with a weak likelihood will have their random effect γ_i shrunk towards the population location. The amount of regularization depends on the variation in the population, a quantity that is estimated from each likelihood. This is how partial pooling works as an adaptive prior for ‘similar’ parameters (**Figure 1B**). The data ‘pools’ some evidence while still allowing for individual differences in samples.

The final term is the exposure E_i . Cell counts may be recorded from sections with different areas. The exposure term scales the parameters in the linear model as the recording area increases ([McElreath, 2018](#)). In our model, the exposure is equal to the logarithm of the recording area; this value is available as part of the experimental data.

The set of parameters $\tau_{r,g}$ models the population standard deviations of the noise for each region r and animal group g . When working on the log scale, priors for these parameters are typically derived in terms of multiplicative increases. Since the parameters are positive they are assigned a half-normal distribution

$$\tau_{r,g} \sim \text{HalfNormal}(\log(s)) \quad (4)$$

with an appropriately chosen scale $s > 1$. For our analyses we used $s = 1.05$ because this gives a HalfNormal distribution with 95% of its density in the interval $[0, \log(1.1)]$. This translates into an approximate 10% variation around $\exp(\theta)$ at the upper end, which is a moderately informative prior, reflecting our belief that within-group animal variability is small relative to between-group variability. This regularization also helps model inference when the datasets are undersampled.

Horseshoe prior

Cell count data often has outliers, for example due to experimental artifacts. Since by default the likelihood does not account for these outliers, they may cause substantial changes in fitted parameter values. This is demonstrated in **Figure 2**, where a careless application of the Poisson distribution on data with several zero counts has a large influence on the posterior distribution. There are two general options for dealing with outliers: either modeling them in the likelihood or in the prior. Although the likelihood option is preferred as it is more direct, see our zero-inflation model below, it can be hard to design because it requires knowledge of the outlier generation process. The alternative is via a flexible prior such as the horseshoe [Carvalho et al. \(2010\)](#); [Piironen and Vehtari \(2017\)](#). This more generic option may be suitable as a default ‘standard’ approach in the typical case where outliers are poorly understood.

Table 1.

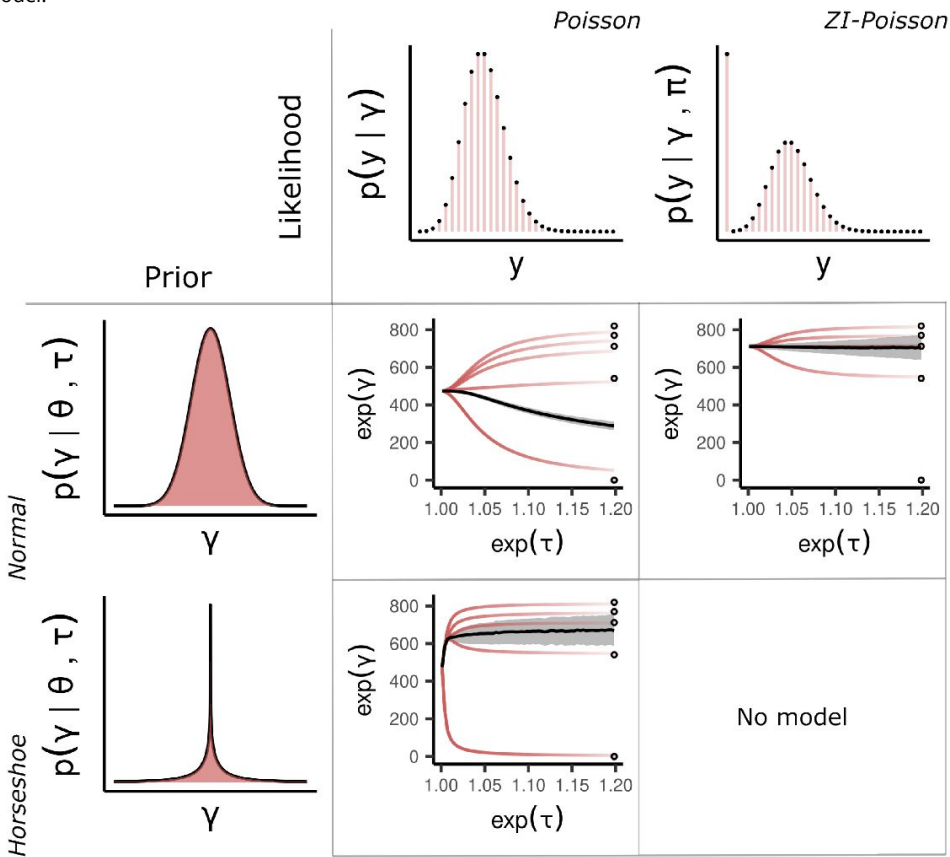
Parameter table for the hierarchical model.

Parameter	Description
E_i	Exposure.
κ_i	Horseshoe inflation.
π	Zero inflation.
γ_i	Random effect for observation i .
θ_{rg}	Fixed effect for region r in group g .
τ_{rg}	Scale of random effects for region r in group g .

Figure 2.

Methods.

A table of partial pooling behaviour for different likelihood and prior combinations. Rows are the two prior choices for the population distribution, and columns the two distributions for the data. Within each cell the expectation of the marginal posterior $p(\exp(\gamma_i) \mid \theta, \tau, y)$ is plotted as a function of τ . The solid black line is the expectation of the marginal posterior $p(\theta \mid \tau, y)$ with one standard deviation highlighted in grey. **Top left:** Combining a normal prior for the population with a Poisson likelihood is unsatisfactory in the presence of a zero observation. The zero observations influence the population mean in an extreme way owing to their high importance under the Poisson likelihood. **Bottom left:** By changing to a horseshoe prior the problematic zero observations can escape the regularization machinery. However, regularization of the estimates with positive observations is much less impactful. **Top right:** A zero-inflated Poisson likelihood accounts for the zero observations with an added process, reducing the burden on the population estimate to compromise between extreme values. **Bottom right:** No model.



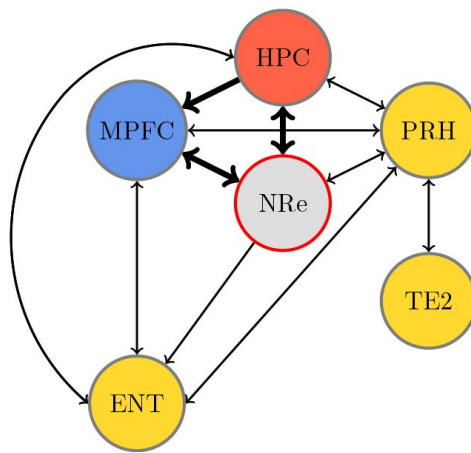


Figure 3.

Recognition memory circuit.

Schematic of the recognition memory network adapted from (Exley, 2019 [Exley, 2019](#)). Bold arrows show the assumed two-way connection between the medial prefrontal cortex and the hippocampus facilitated by the NRe. Colours highlight the HPC (red), MPC (blue) and specific areas of the rhinal cortex (yellow). The NRe was lesioned in the experiment.

The horseshoe prior is a hierarchical prior for sparsity. It introduces an auxiliary parameter κ_i that multiplies the population scale τ . This construction allows surprising observations far from the bulk of the population density to escape regularisation.

$$\gamma_i \sim \text{Normal}(0, \tau_{r[i],g[i]} \times \kappa_i) \quad (5)$$

$$\kappa_i \sim \text{HalfNormal}(1). \quad (6)$$

An example of this is given in [Figure 2](#) as the bottom left cell of the 2×2 table of models. The horseshoe prior often uses a Cauchy distribution but in our case the heavy tail causes problems for the sampling algorithm, see SI section 8.

Zero inflation

A particular trait of the second dataset is that there are a large number of zero data points ($\sim 6\%$). Although a zero observation is always possible for a Poisson distribution, for plausible values of the propensity, zeros should be rare. It is likely that for some regions the experiment has not worked as expected and the zeros show that something has ‘gone wrong’ and that the readings are not well described by a Poisson distribution. Here we extend the model to include this possibility. This is a useful elaboration of the standard model. In the standard model the horseshoe prior ensures that these anomalous readings only have a small effect on the result but it is more informative to extend the model to include them. While this particular extension is specific to these data, it also serves as an example of how a standard Bayesian model can serve as a starting point for an iterative investigation of the data.

The zero-inflated Poisson model is intended to model a situation where there are zeros unrelated to the Poisson distribution, in this case this might, for example, be the result of an error in the automated registration process that identifies regions and counts their cells. It is a mixture model, if

$$y_i \sim \text{ZIPoisson}(\pi, \lambda_i) \quad (7)$$

there is a probability π that $y_i = 0$ and a $1 - \pi$ probability that y_i follows a Poisson distribution with rate λ_i . Importantly this means there are two ways in which y_i can be zero, through the Bernoulli process parameterized by π , or through the Poisson distribution. This has the effect of ‘inflating’ the probability mass at zero with the additional parameter π giving the proportion of extra zeros in the data that could not be explained by the standard Poisson distribution. This distribution can be visualized in [Figure 2](#) and further mathematical details are described in SI 2C.

Model inference

Posterior inference was performed with the probabilistic programming language Stan ([Carpenter et al., 2017](#)), using its custom implementation of the No-U-Turn (NUTS) sampler [Betancourt \(2016\)](#); [Hoffman et al. \(2014\)](#). For each model, the posterior was sampled using four chains for 8000 iterations, with half of these being attributed to the warm-up phase. This gives a total of 16000 samples from the posterior distribution.

Results

We describe differences in estimated counts between groups in terms of $\log_2$ -fold changes. Fold changes are useful because they prevent differences that are small in absolute magnitude from being masked by regions with high overall expression. Our results compare Bayesian highest density intervals (HDI) with the confidence interval (CI) from an uncorrected Welch’s t -test. The

Bayesian HDI is calculated from the posterior distribution and is the smallest width interval that includes a chosen probability, here 0.95 (to correspond to $\alpha = 0.05$), and summarizes the meaningful uncertainty over a parameter of interest.

Case study one - transient neural activity in the recognition memory circuit

Results for the first dataset are presented in **Figure 4**. **Figure 4A** plots cell-count differences between the novel and familiar conditions without lesion and **Figure 4B** with lesion. These data were collected to investigate the role of different hippocampal and adjacent cortical regions in memory. However, some regions of interest, such as the intermediate dentate gyrus (IDG), and the dorsal subiculum (DSUB) look under-powered: for both regions there is a markedly non-zero difference in expression between the novel and familiar conditions in the sham animals but a wide confidence interval overlapping zero makes the evidence unreliable (orange bars **Figure 4A**).

In contrast, the Bayesian estimates (green bars **Figure 4**) produce a clear result. For a number of brain regions in **Figure 4A**, sham-novel animals have higher expression than sham-familiar ones. These differences disappear in **Figure 4B** with lesion-novel and lesion-familiar animals showing roughly equal cell counts. This indicates that the difference is only present when the NRe is intact.

Case study two - Ontogeny of inhibitory interneurons of the mouse thalamus

For each of the fifty brain regions the estimated $\log_2$ -fold difference in GFP expressing cells between the two genotypes is plotted in **Figure 5**. This includes the purple and pink 95% HDI from the horseshoe and zero-inflated Poisson models along with the 95% confidence interval arising from a t -test in orange. For most brain regions, the two Bayesian models gave narrower HDIs than the t -test CI. Accordingly, the Bayesian models identified a greater number of brain regions that had genotype differences in *Sox14*-positive cell count in the sense that it found more places where the appropriate uncertainty interval does not overlap zero.

Despite the large difference in interval estimation between the Bayesian HDIs and t -test CI for many brain regions, as the data becomes stronger from the perspective of the frequentist p -value towards the right-hand side of the second row in **Figure 5**, the model results become much more compatible. The variation within groups is very small for these regions. Further regularization is not necessary and so the impact of partial pooling has been reduced. The sample estimate of the t -test has ‘caught up’ to the regularized estimate because the signal is strong.

The zero-inflated Poisson distribution sometimes differs with the t -test confidence intervals. One example of this is the result for dorsal tuberomammillary nucleus (TMD). **Figure 6**, bottom row, plots the raw cell count values for TMD alongside the inferred frequentist mean and two Bayesian model means. For this region the HET animals have high GFP expression across both hemispheres, yet animal three has a reading of zero for both hemispheres. This injects variability into the standard deviation of the HET group. Consequently, the pooled standard deviation used in the t -test is large and almost certainly guarantees a non-significant result. Furthermore, the sample mean of this region looks nothing like zero, but also nothing like the other two animals with positive counts. In addition to the wide interval, the sign of the difference does not agree with the data. The medial preoptic nucleus (MPN) also suffers from poor estimation. Once again, this region contains a single HET animal for which the reading from both hemispheres is zero. The zero-inflated Poisson produces a posterior distribution of the appropriate sign with small uncertainty.

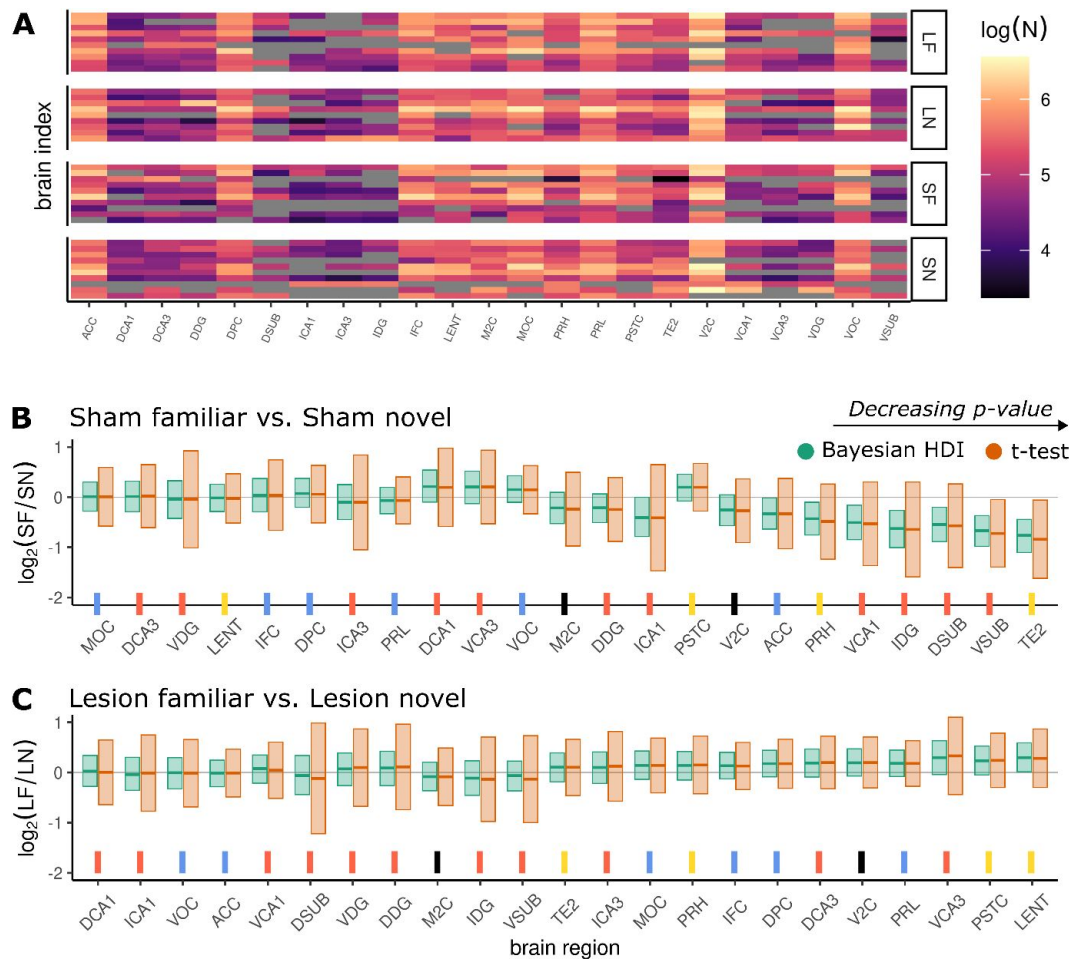


Figure 4.

Results—Case study 1.

A: Heatmap of the raw log cell count data. Each row corresponds to a single animal, columns correspond to brain regions. Animals are grouped into lesion-familiar (LF), lesion-novel (LN), sham-familiar (SF) and sham-novel (SN). **B,C:** $\log_2$ -fold differences for each surgery type: B shows differences between SF and SN groups, C shows differences between LF and LN groups. The 95% Bayesian HDI is given in green, and the 95% confidence interval calculated from a Welch's t -test in orange. Horizontal lines within the intervals mark the posterior mean of the Bayesian results, and the raw data means in the t -test case. The x-axis is ordered in terms of decreasing p -value from the significance test and ticks have been color-paired with the nodes in the recognition memory circuit diagram [Figure 3](#). Black ticks are not present in the circuit because they are the control regions in the experiment.

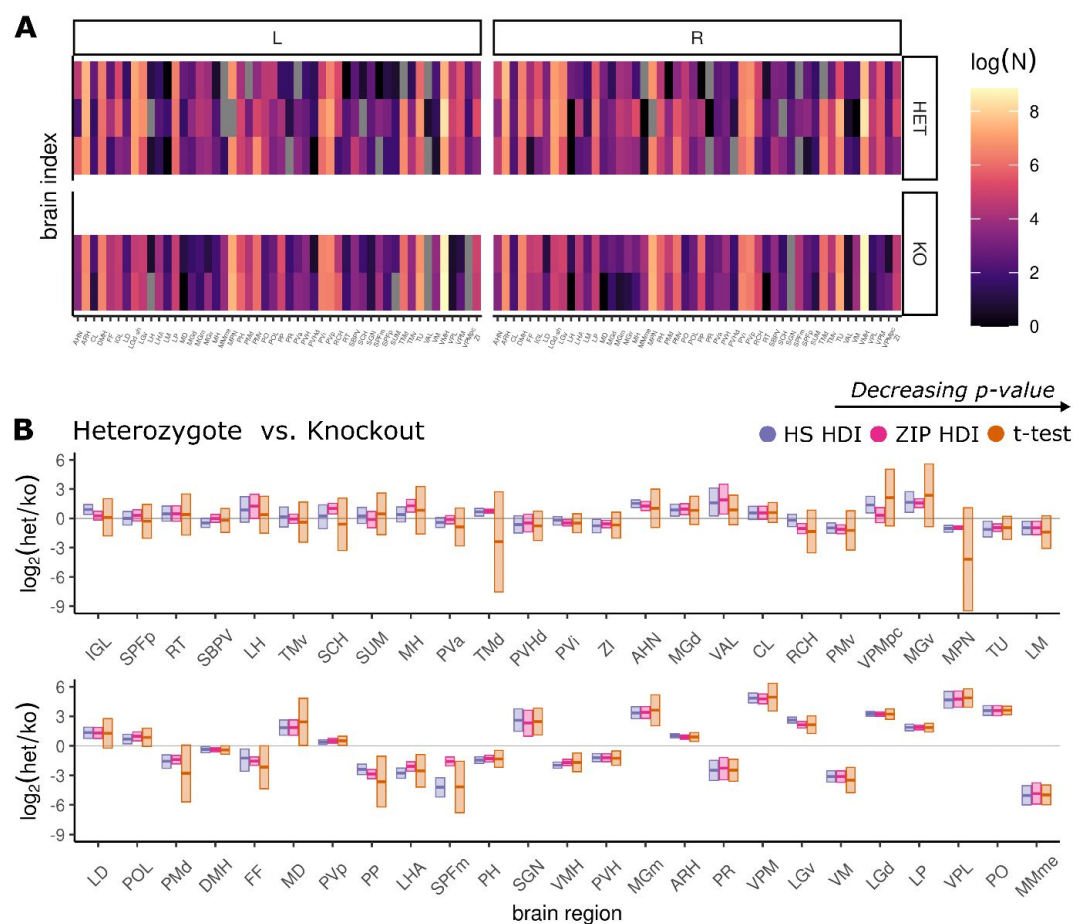


Figure 5.

Results—Case study 2.

A: Heatmap of the raw log cell count data. Each row corresponds to a single animal, columns correspond to brain regions. L and R denote left and right hemispheres, respectively. **B:** $\log_2$ fold differences in GFP positive cells between mouse genotypes, heterozygous (HET) and knockout (KO), for each of the fifty recorded brain regions spread across two rows. The 95% Bayesian HDI is given in purple and pink for the Bayesian horseshoe and zero-inflated model. The 95% confidence interval calculated from a Welch's t -test in orange. Horizontal lines within the intervals mark the posterior mean of the Bayesian results and the data estimate for the t -test. The x -axis is ordered in terms of decreasing p -value from the significance test.

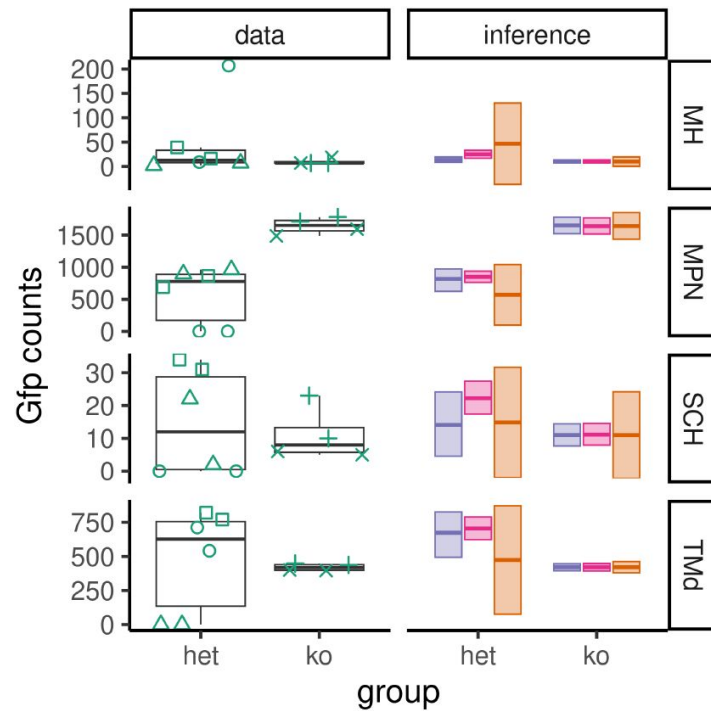


Figure 6.

Example data and inferences highlighting model discrepancies.

On the left under 'data': boxplots with medians and interquartile ranges for the raw data for four example brain regions. The shape of each point pairs left and right hemisphere readings in each of the five animals. On the right under 'inference': HDIs and confidence intervals are plotted. Purple is the Bayesian horseshoe model, pink is the Bayesian ZIP model, and orange is the sample mean. The Bayesian estimates are not strongly influenced by the zero-valued observations (MPN, SCH, TMd) or large-valued outliers (MH) and have means close to the data median. This explains the advantage of the Bayesian results over the confidence interval.

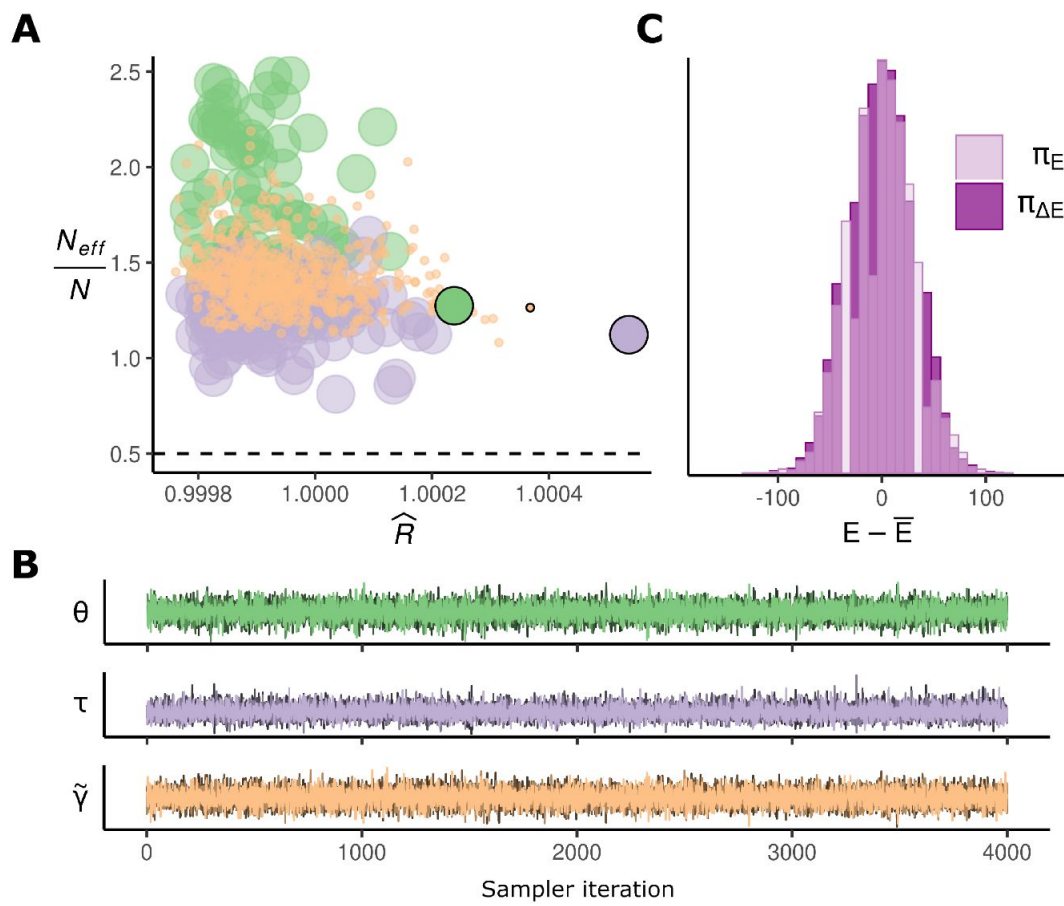


Figure 7.

Diagnostics - Poisson.

Standard Poisson model, case study - 1.

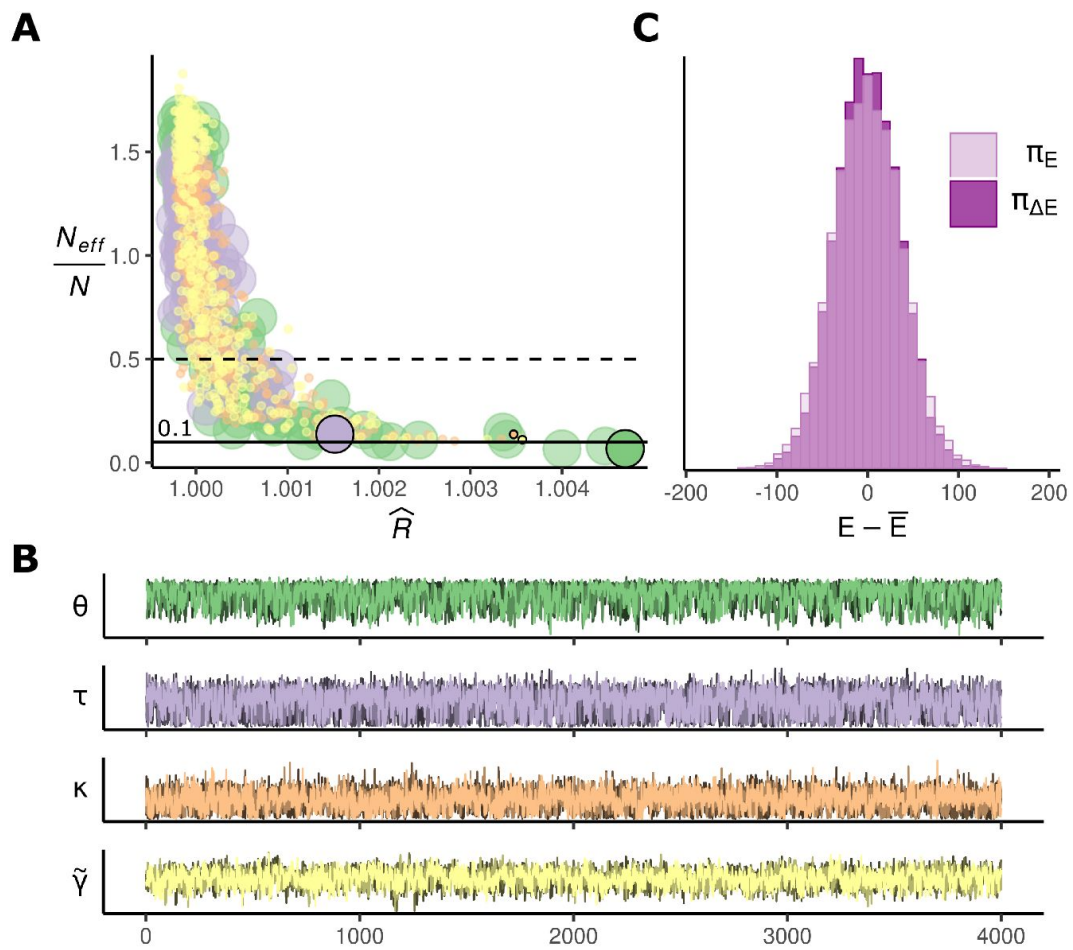


Figure 8.

Diagnostics - Horseshoe.

Horseshoe model, case study - 2.

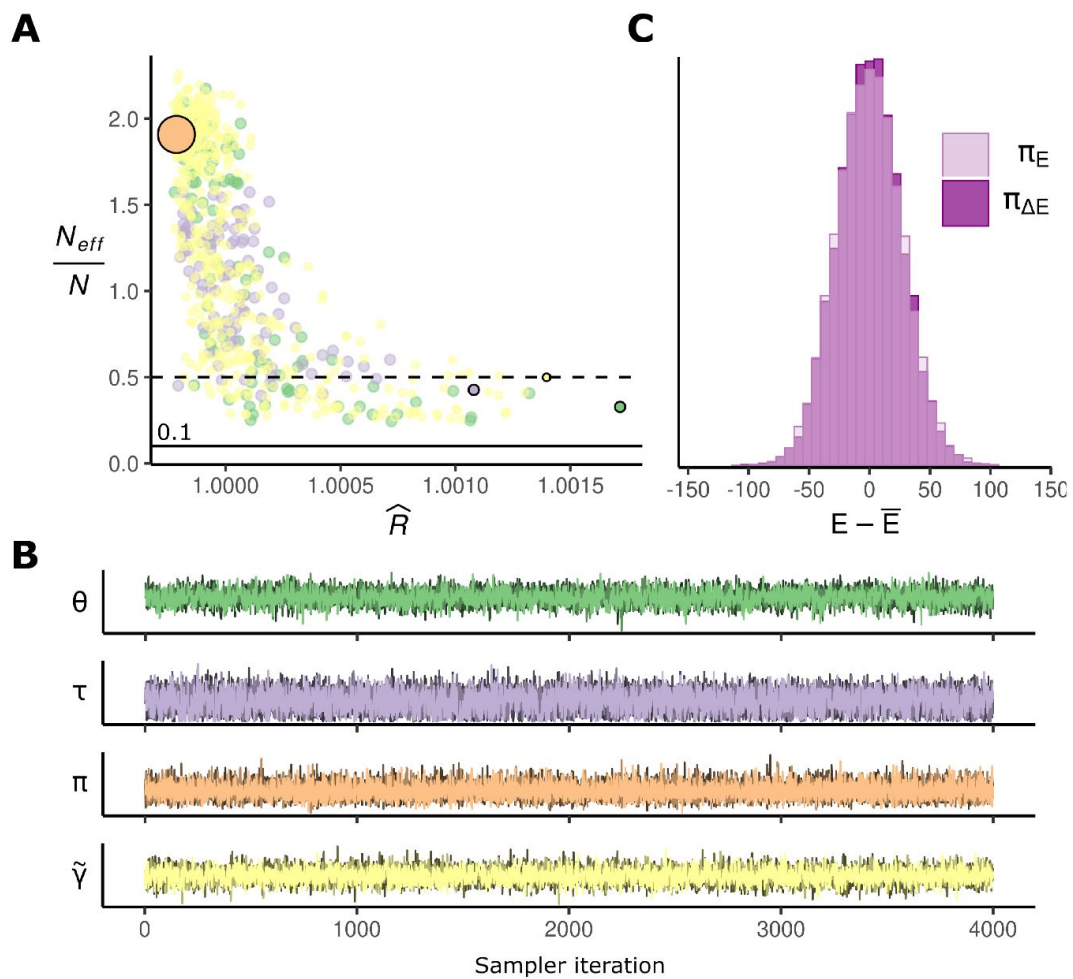


Figure 9.

Diagnosics - ZIPoisson.

Zero-inflated Poisson, case study - 2.

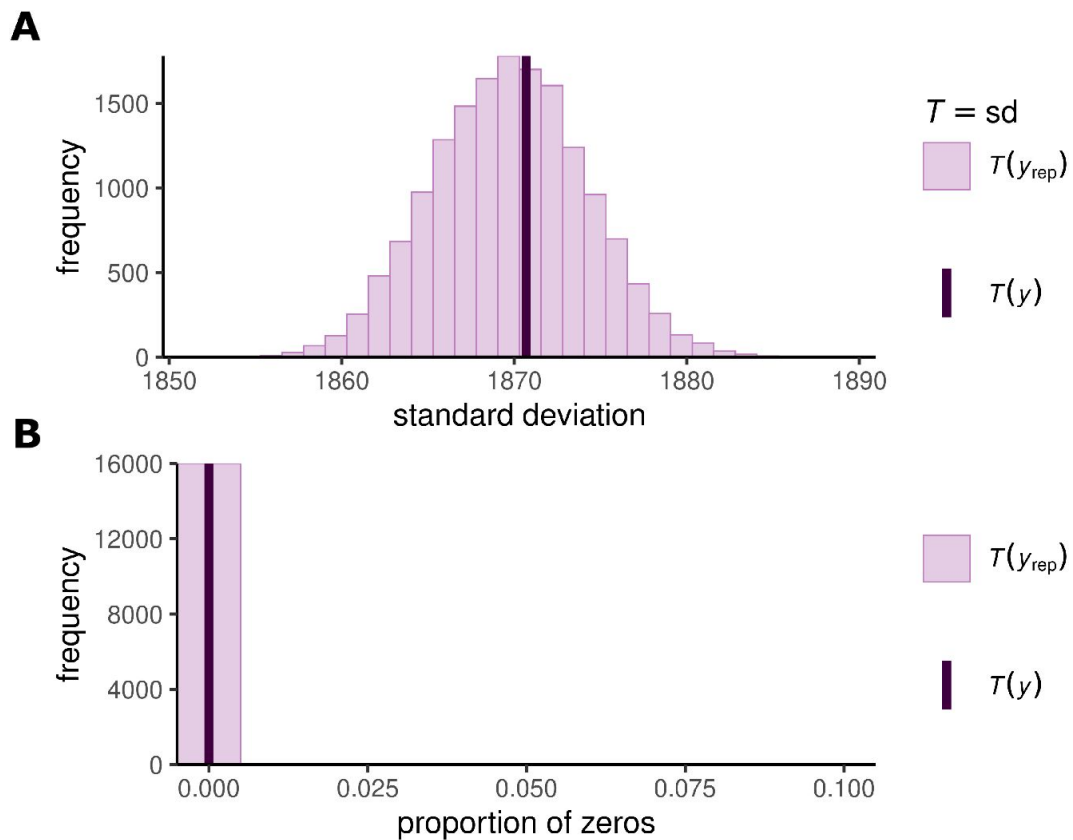


Figure 10.

PPC - Poisson.

Posterior predictive check for the standard Poisson model in case study - 1. **A:** The proportion of zeroes in the data matches the proportion of zeroes in posterior predictive samples. This proportion is zero. **B:** The distribution of standard deviations computed over a number of posterior predictive datasets (histogram), aligns with the standard deviation of the data.

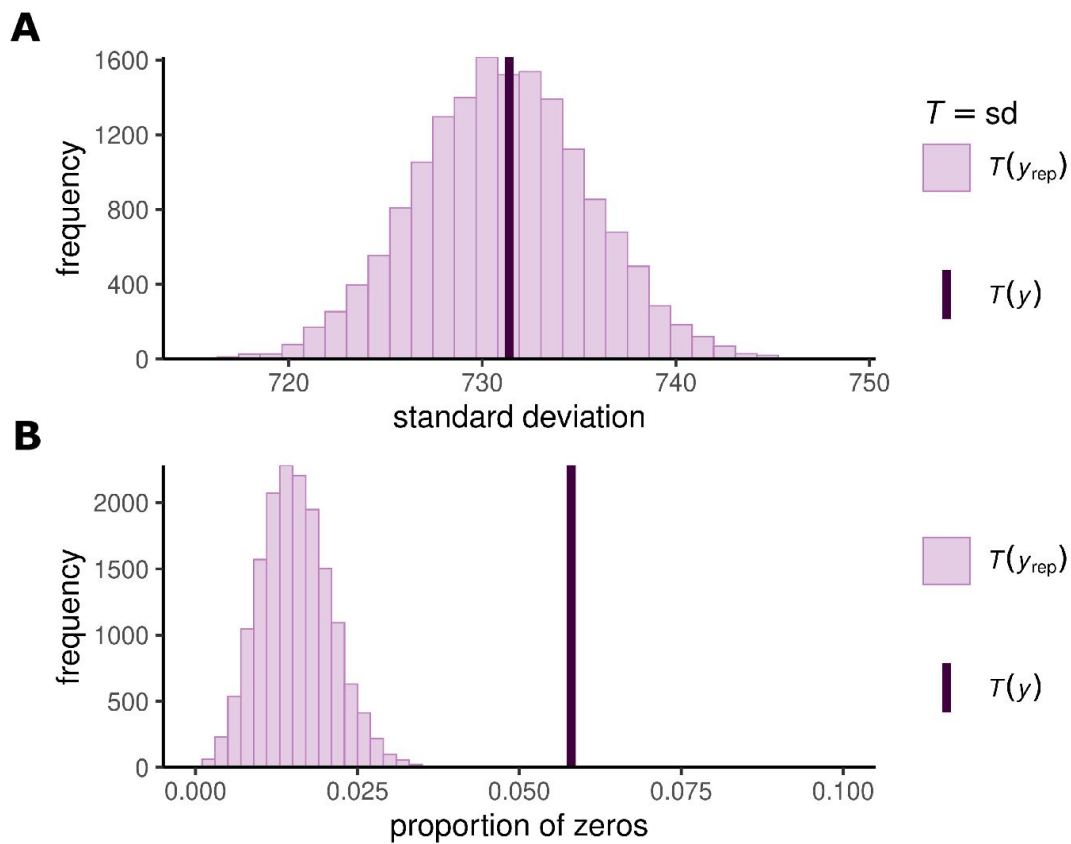


Figure 11.

PPC - Horseshoe.

Horseshoe model, case study - 2. Posterior predictive check for the standard horseshoe model in case study - 2. **A:** The proportion of zeroes in the data is larger than those found in posterior predictive datasets. This makes sense, because the likelihood is still a Poisson distribution. **B:** The distribution of standard deviations computed over a number of posterior predictive datasets (histogram), aligns with the standard deviation of the data.

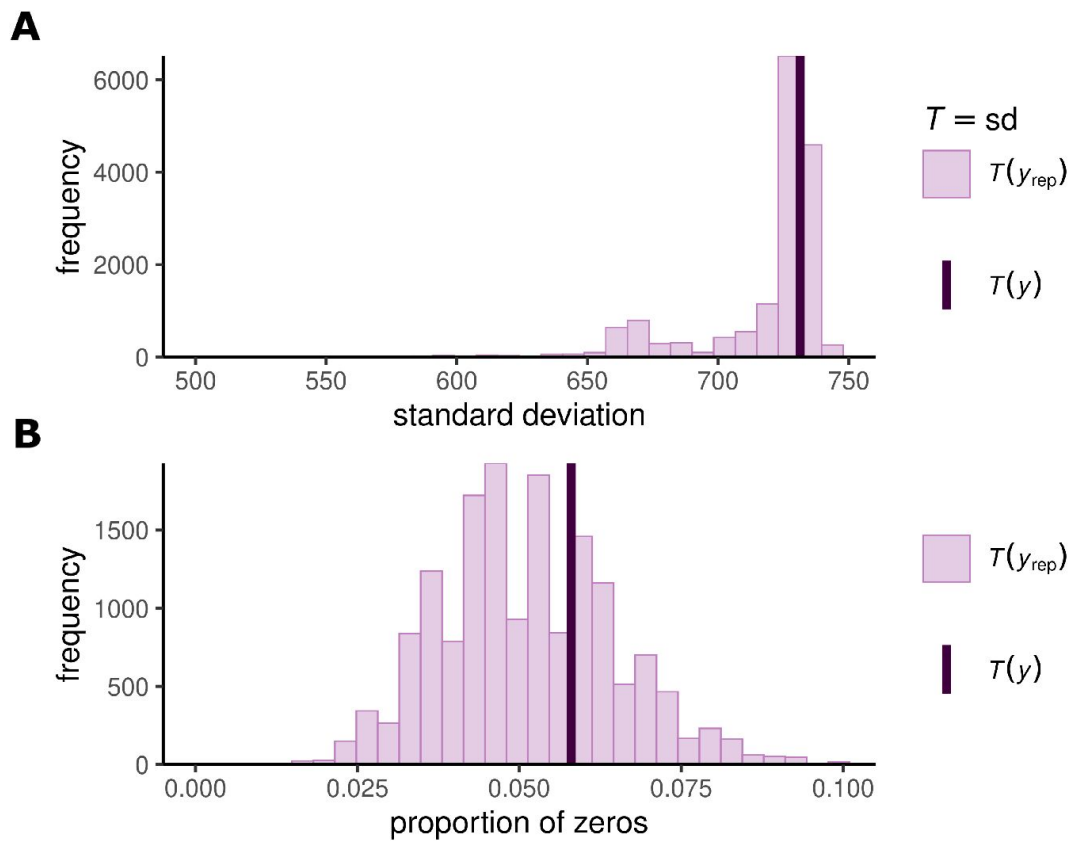


Figure 12.

PPC - ZIPoisson.

Zero-inflated Poisson, case study - 2. **A:** The proportion of zeroes in the data matches the proportion of zeroes in posterior predictive samples. **B:** The distribution of standard deviations computed over a number of posterior predictive datasets (histogram), aligns with the standard deviation of the data.

The two Bayesian models did not always agree. In some cases, such as the Medial habenula (MH) and Suprachiasmatic nucleus (SCH), the ‘standard’ horseshoe model does not show a genotype difference in cell counts, while the ZIP model indicates that heterozygotes had higher cell counts than knock-outs ([Figure 5](#), [Figure 6](#)). The opposite can be seen in the case of the parvicellular ventral posteromedial nucleus of the thalamus (VPMpc), the horseshoe model suggests a genotype difference where the ZIP model did not ([Figure 5](#)). Further examination of the data shows why this happens, for example for region MH ([Figure 6](#), top row), the ‘standard’ horseshoe model sensibly ignores the large positive outlier value in the heterozygote data, while the ZIP model does not. As a result the ZIP model’s estimate for the mean is pulled upward, leading to an inferred difference in heterozygote versus the wild-type.

Discussion

We have presented a standard workflow for Bayesian analysis of multi-region cell-count data. We propose a likelihood and appropriate priors with a nested hierarchical structure reflecting the structure of the experiment. We applied this to two distinct example datasets and demonstrate that they capture more fruitfully the characteristics of the data when compared to field-standard frequentist analyses.

For both case studies, the Bayesian uncertainty intervals are more precise than the confidence intervals. These confidence intervals tend to be quite wide on these data because of the small sample size and because of violations to their parametric model assumptions. Our standard workflow uses a horseshoe prior, along with the partial pooling, this allows our model to deal effectively with outliers. Furthermore, for the data sizes presented here, a full Bayesian inference using Stan does not require long computation time, or even particularly high performance hardware. Modern multi-core laptop processors are quite sufficient for this task. Fitting a model typically takes less than an hour.

In our analysis we have noted examples where different Bayesian models give discrepant conclusions. The obvious question to ask is which should we trust? The disappointing but inevitable answer is that, as with more traditional methods, Bayesian analysis is only a tool useful for interpreting data and brings with it a set of assumptions and biases regarding the experiment and the data. A Bayesian analysis does not avoid inconsistent or inconclusive results but it usually makes the assumptions more explicit and transparent. Typically the solution these model inconsistencies is to inspect the raw data and ask which model better captures those aspects of the data we are most interested in. Overall, the lesson here is that Bayesian hierarchical modeling has greater flexibility and statistical power but all statistical analyses, even those claiming to ‘test hypotheses’, just support exploration and it is ultimately the researcher’s responsibility to make sure that a model’s assumptions are appropriate and its behavior is sensible for the target dataset.

The horseshoe prior model workflow we have exhibited here is intended as a standard approach. We believe that, without extension, it will provide a robust model for cell-count data. However, we also suggest that the standard workflow can be a useful first step for a more comprehensive, extended, model when one is required. We have given an example of this for the second dataset where the anomalous zeros prompted us to change the likelihood to a zero-inflated Poisson. There are other possibilities, for example, zero inflation is not the only way to handle an anomaly in the number of zeros: the hurdle model is an alternative, [Cragg \(1971\)](#). This is not a mixture model, instead it restricts the probability of zeros to some value π with the probabilities for the positive counts coming from a truncated Poisson distribution. The hurdle model can deflate as well as inflate the probability mass at zero. This did not match the situation in the data we considered but might for other datasets. Another extension might involve tighter priors based on previous

experiments. This is likely to be very relevant for cell-count data since these experiments are rarely performed in isolation and so prior information can be leveraged from a history of empirical results.

One obvious elaboration of our model would replace normal distributions with multivariate normal distributions. This would have two advantages. Firstly, correlations are difficult to estimate for under-sampled data. Including correlation matrix priors provides extra information, for example, based on anatomical connectivity, that can aid the statistical estimation of other parameters. Secondly, it would more closely match our understanding of the experiment: we know that activity is likely to be correlated across regions and so it is apposite to include that directly in the model. Unfortunately the problem of finding a suitable prior for the correlation proved insurmountable: the standard Lewandowski-Kurowicka-Joe distribution [Lewandowski et al. \(2009\)](#) which has been useful in lower-dimensional situations is too regularizing here. This is an area where further work needs to be done.

It is important to highlight that a mixed effects model is not a uniquely Bayesian construction. Indeed, any model that tries to include more sophistication through hierarchical structures, Bayesian or otherwise, is useful. However, non-Bayesian models can be complicated and opaque, they are also often more restrictive, for example they often assume normal distributions, and circumventing these restrictions can make the models even less transparent. A Bayesian approach is, at first, unfamiliar; this can make it seem more obscure than better established methods, but, in the long run, Bayesian models are typically clearer and do not involve so many different assumptions and so many fine adjustments.

Code and data availability

The code necessary to run the models presented in this manuscript can be found at [Dimmock et al. \(2025\)](#) and on our Github <https://BayesianCellCounts.github.io>. The data for case study one on nucleus reuniens lesion are available from <https://doi.org/10.5281/zenodo.12787211> [Exley et al. \(2024\)](#). The data from case study two on Sox14 expressing neurons are available from <https://doi.org/10.5281/zenodo.12787287> [Gerald et al. \(2024\)](#).

Appendix 1

Full model

Here we present the complete mathematical model for each of the three models applied in the main text. In all cases, the exposure term is only necessary for the first case study. Area was not available for the second so the exposure term was left out.

Poisson model

$$y_i \sim \text{Poisson}(\lambda_i) \quad (8)$$

$$\log \lambda_i = E_i + \gamma_i \quad (9)$$

$$\theta_{rg} \sim \text{Normal}(5, 2) \quad (10)$$

$$\gamma_i \sim \text{Normal}(\theta_{r[i],g[i]}, \tau_{r[i],g[i]}) \quad (11)$$

$$\tau_{rg} \sim \text{HalfNormal}(\log(1.05)) \quad (12)$$

Horseshoe model

$$y_i \sim \text{Poisson}(\lambda_i) \quad (13)$$

$$\log \lambda_i = E_i + \gamma_i \quad (14)$$

$$\theta_{rg} \sim \text{Normal}(5, 2) \quad (15)$$

$$\gamma_i \sim \text{Normal}(\theta_{r[i],g[i]}, \kappa_i \times \tau_{r[i],g[i]}) \quad (16)$$

$$\tau_{rg} \sim \text{HalfNormal}(\log(1.05)) \quad (17)$$

$$\kappa_i \sim \text{HalfNormal}(1) \quad (18)$$

Zero-inflated model

$$y_i \sim \text{ZIPoisson}(\lambda_i, \pi) \quad (19)$$

$$\pi \sim \text{Beta}(1, 5) \quad (20)$$

$$\log \lambda_i = E_i + \gamma_i \quad (21)$$

$$\theta_{rg} \sim \text{Normal}(5, 2) \quad (22)$$

$$\gamma_i \sim \text{Normal}(\theta_{r[i],g[i]}, \tau_{r[i],g[i]}) \quad (23)$$

$$\tau_{rg} \sim \text{HalfNormal}(\log(1.05)) \quad (24)$$

Distributions

Zero inflated Poisson distribution

The zero inflated Poisson distribution is a mixture distribution with mixing parameter π . The distribution is formally defined below. If,

$$p_Y(y) \equiv \text{ZIPoisson}(\pi, \lambda) \quad (25)$$

Then

$$p_Y(y) = \begin{cases} \pi + (1 - \pi) \exp(-\lambda) & \text{if } y = 0 \\ (1 - \pi)f(y) & \text{otherwise} \end{cases} \quad (26)$$

Where

$$f(y) \equiv \text{Poisson}(\lambda) \quad (27)$$

Equivalently, the mixture can be specified with an indicator function.

$$p_Y(y) = \mathbf{1}_0(y)\pi + (1 - \pi)f(y) \quad (28)$$

Half Normal distribution

The HalfNormal distribution coincides with a zero-mean normal distribution truncated at zero. It has a single scale parameter σ . If,

$$p_Y(y) \equiv \text{HalfNormal}(\sigma) \quad (29)$$

Then

$$p_Y(y) = \frac{\sqrt{2}}{\sigma\pi} \exp\left(-\frac{y^2}{2\sigma^2}\right) \quad (30)$$

is the probability density function.

Additional methods

Fold differences

In our results we present differences between experimental groups in terms of $\log_2$ -fold differences. We calculate this as follows. The parameter of interest $\theta_{r,g=i}$ is modeled on the natural log scale owing to the log-link function necessary for the Poisson regression. At the average animal the difference,

$$\theta_{r,g=i} - \theta_{r,g=j} = \log \frac{\exp(\theta_{r,g=i})}{\exp(\theta_{r,g=j})} \quad (31)$$

$$= \log \Delta_{ij} \quad (32)$$

is the natural log of the ratio of the expected counts. To obtain $\log_2$ -fold differences we simply change the base by multiplying $\log \Delta_{ij}$ by $\log_2(e)$.

Data transformations

To facilitate a comparison with the Bayesian intervals in terms of $\log_2$ -fold differences, it was necessary to add one to any zero counts before applying the t -test.

Non-centered parameterisation

Hierarchical models can produce geometry that is difficult for the sampler to explore. Fortunately, there exists a simple reparameterisation known as non-centering that can remedy this problem. In our model, instead of sampling y_i directly, we sample the parameter $\tilde{y}_i$ instead, and use it to reconstruct y_i . That is, sample $\tilde{y}_i$ from a standard normal distribution,

$$\tilde{y}_i \sim \text{Normal}(0, 1) \quad (33)$$

and reconstruct y_i as a deterministic function of sampled values of $\tilde{y}_i$ and $\tau_{r[i],g[i]}$.

$$y_i = \theta_{r[i],g[i]} + \tau_{r[i],g[i]} \times \tilde{y}_i \quad (34)$$

This removes the frustrating joint behaviour between of $(y_i, \tau_{r[i],g[i]})$, and promotes efficient sampling.

Preprocessing - case study 1

In these data some animals produced more than one reading per brain region. Before fitting our model these were summed together to produce a single count; the exposure term was also properly adjusted to correctly reflect the area of the recording site.

Software, packages and libraries

R Libraries	Version	Description
rstan	2.26.3	complete Stan library
cmdstanr	0.5.2	lightweight Stan library
HDInterval	0.2.2	calculating HDI in R
ggplot2	3.4.1	plotting
bayesplot	1.9.0	plotting
tidyverse	1.3.1	tibble, tidyr, readr, purr, dplyr, stringr, forcats

- R version 4.2.1–“Funny-looking-kid”.
- Computation was performed locally on a Dell XPS 13 7390 laptop [Intel i7-10510U @ 1.80 GHz, 16 GB of RAM; Ubuntu 20.04.4 LTS].
- Panels composed using inkscape version 1.2.2.

Brain-region name acronyms

Here we include a list of acronyms for the brain regions in each case study.

Term	Definition
ACC	Anterior cingulate cortex.
DCA1/3	Dorsal CA1/3
DDG	Dorsal dentate gyrus.
DPC	Dorsal peduncular cortex.
DSUB	Dorsal subiculum.
HPC	hippocampus.
ICA1/3	Intermediate CA1/3
IDG	Intermediate dentate gyrus.
IFC	Infralimbic cortex.
LENT	Lateral entorhinal cortex.
MOC	medial orbital cortex.
MPFC	Medial prefrontal cortex
M2C	Motor cortex M2
NRe	Nucleus reuniens
PRL	prelimbic cortex.
PRH	Perirhinal cortex
PSTC	postrhinal cortex.
TE2	Temporal association cortex
VCA1/3	Ventral CA1/3
VDG	Ventral dentate gyrus.
VOC	Ventral orbital cortex.
VSUB	Ventral subiculum.
V2C	Visual cortex V2

Term	Definition	Term	Definition
AHN	Anterior hypothalamic nucleus	PP	Peripeduncular nucleus
ARH	Arcuate hypothalamic nucleus	PR	Perireunensis nucleus
CL	Central lateral nucleus of the thalamus	PVa	Periventricular hypothalamic nucleus, anterior part
DMH	Dorsomedial nucleus of the hypothalamus	PVH	Paraventricular hypothalamic nucleus
FF	Fields of Forel	PVHd	Paraventricular hypothalamic nucleus, descending division
IGL	Intergeniculate leaflet of the lateral geniculate complex	PVi	Periventricular hypothalamic nucleus, intermediate part
LD	Lateral dorsal nucleus of thalamus	PVp	Periventricular hypothalamic nucleus, posterior part
LM	Lateral mammillary nucleus	RCH	Retrochiasmatic area
LGv	Ventral part of the lateral geniculate complex	RT	Reticular nucleus of the thalamus
LGd	Dorsal part of the lateral geniculate complex	SBPV	Subparaventricular zone
LH	Lateral habenula	SCH	Suprachiasmatic nucleus
LHA	Lateral hypothalamic area	SGN	Suprageniculate nucleus
LP	Lateral posterior nucleus of the thalamus	SPFm	Subparafascicular nucleus, magnocellular part
MD	Mediodorsal nucleus of thalamus	SPFp	Subparafascicular nucleus, parvicellular part
MGd	Medial geniculate complex, dorsal part	SUM	Supramammillary nucleus
MGv	Medial geniculate complex, ventral part	TMd	Tuberomammillary nucleus, dorsal part
MGm	Medial geniculate complex, medial part	TMv	Tuberomammillary nucleus, ventral part
MH	Medial habenula	TU	Tuberal nucleus
MMme	Medial mammillary nucleus, median part	VAL	Ventral anterior-lateral complex of the thalamus
MPN	Medial preoptic nucleus	VMH	Ventromedial hypothalamic nucleus
PH	Posterior hypothalamic nucleus	VM	Ventral medial nucleus of the thalamus
PMd	Dorsal premammillary nucleus	VPL	Ventral posterolateral nucleus of the thalamus
PMv	Ventral premammillary nucleus	VPM	Ventral posteromedial nucleus of the thalamus
PO	Posterior complex of the thalamus	VPMpc	Ventral posteromedial nucleus of the thalamus, parvicellular part
POL	Posterior limiting nucleus of the thalamus	ZI	Zona incerta

Sampler diagnostics

The basic Poisson model (Equation 1 [↗](#)) was sampled excellently. Measures of sampling performance such as $\hat{R}$ Gelman and Rubin (1992 [↗](#)); Vehtari et al. (2021 [↗](#)), and effective sample size, were all satisfactory (SI figure 1). Similarly, for the zero-inflated model (Equation 7 [↗](#)), no problems were observed for any of the diagnostics (SI figure 2). Contrasting with this, the horseshoe model (Equation 6 [↗](#)), exhibited some signs of fitting problems (SI figure 3). Divergences were not observed, and given the longer chain length this is reassuring evidence against biased computation. However for many parameters the effective sample size is much lower than we would like to see. This is reflected in the trace plots: the sampler is not making large jumps across the parameter space implying high auto-correlation and low effective sample size. Unfortunately, the horseshoe is notoriously hard to fit and we resort to brute-force methods such

as increasing the number of iteration and reducing the step size of the sampler to improve the inference. In the following three plots diagnostics have been summarised with the following three items:

- **A:** The performance of the sampler is illustrated by plotting $\hat{R}$ (R-hat, $\hat{R} \approx 1$ ideal) against the ratio of the effective number of samples (larger is better) for each parameter in the model. Points represent individual parameters in the model, and have further been colour coded by their type. For example, all $\theta_{r,g}$ are coloured in green. Points have also been scaled based on how numerous the parameters are so the more numerous parameters have smaller dots, the less numerous, larger.
- **B:** A histogram comparing the marginal energy distribution π_E , and the transitional energy distribution $\pi_{\Delta E}$ of the Hamiltonian. Ideally, these distributions should match each other closely if the posterior distribution has been properly explored by the sampler.
- **C:** For each parameter type the parameter with the ‘poorest’ mixing (largest $\hat{R}$) are presented with a post-warmup trace plot that overlays the ordered sequence of samples from each of the four chains. Corresponding points in **A** are marked with a black border and zero transparency.

Posterior predictive checking

Posterior predictive checks use the posterior predictive distribution,

$$p(y^{\text{rep}}|y) = \int p(y^{\text{rep}}|\theta)p(\theta|y) d\theta \quad (35)$$

where $p(\theta|y)$ is the posterior distribution, and $p(y^{\text{rep}}|\theta)$ is the data distribution for y^{rep} that follows the same form as the likelihood for y . If we can verify that the posterior predictive distribution can generate replicate datasets with similar statistics to the observed data, then we might conclude that our model is consistent with the observed data and useful for answering questions about it. In practice a Monte-carlo approach is used to approximate statistics of the posterior predictive distribution. For example, if T is the test statistic of interest such as the sample mean, then

1. For $1, \dots, S$.
 1. sample $\theta_s \sim p(\theta|y)$
 2. sample $y_s^{\text{rep}} \sim p(y^{\text{rep}}|\theta_s)$
 3. calculate $T(y_s^{\text{rep}})$, where T is the statistic of interest
2. return $\{T(y_1^{\text{rep}}), \dots, T(y_S^{\text{rep}})\} \approx p(T(y^{\text{rep}})|y)$

The posterior predictive checks that follow examine two important statistics of count data. 1) The standard deviation of the data (dispersion), as figure **a**. 2) The proportion of zeroes in the data (zero-inflation), as figure **B**. The value of the test statistic applied to the observed data $T(y)$ is plotted as a solid purple line, and the distribution of test statistics $\{T(y_1^{\text{rep}}), \dots, T(y_S^{\text{rep}})\}$ as a purple histogram.

Horseshoe densities

In our model a horseshoe prior was used to allow some y_i , typically those informed by $y_i = 0$, to escape regularisation by partial pooling. However, we encountered many problems with the default parameterisation that assigns a HalfCauchy density to the individual inflation parameters κ_i . In **Figure 13A**, the proportional conditional posterior density

$$p(\tilde{y}_i, \kappa_i | \theta, \tau, y_i) \propto p(y_i | \tilde{y}_i, \theta, \tau, \kappa_i) p(\tilde{y}_i) p(\kappa_i) \quad (36)$$

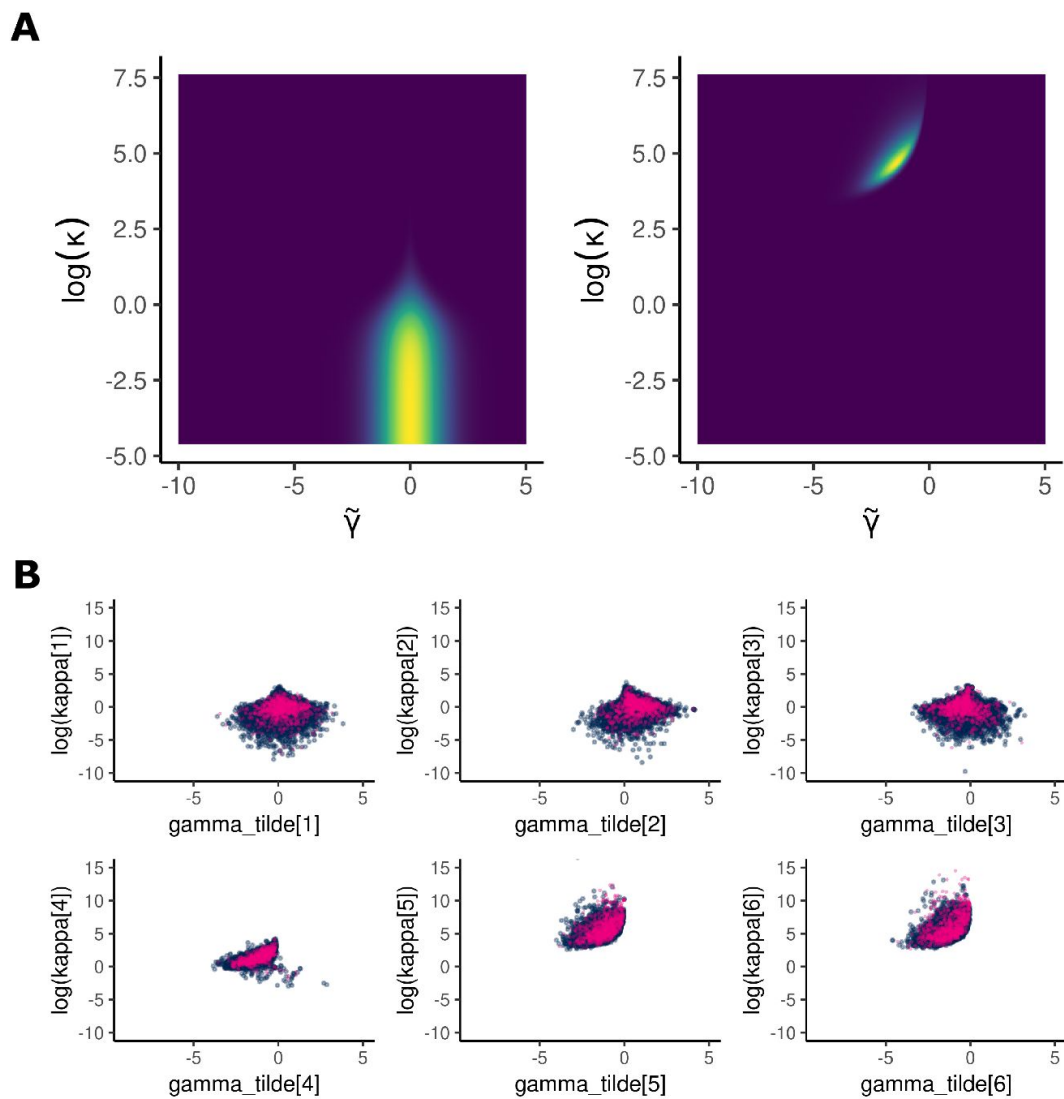


Figure 13.

Horseshoe densities.

A: Conditional posterior. **B:** MCMC pair plots. Divergent samples are coloured in pink, non-divergent in blue.

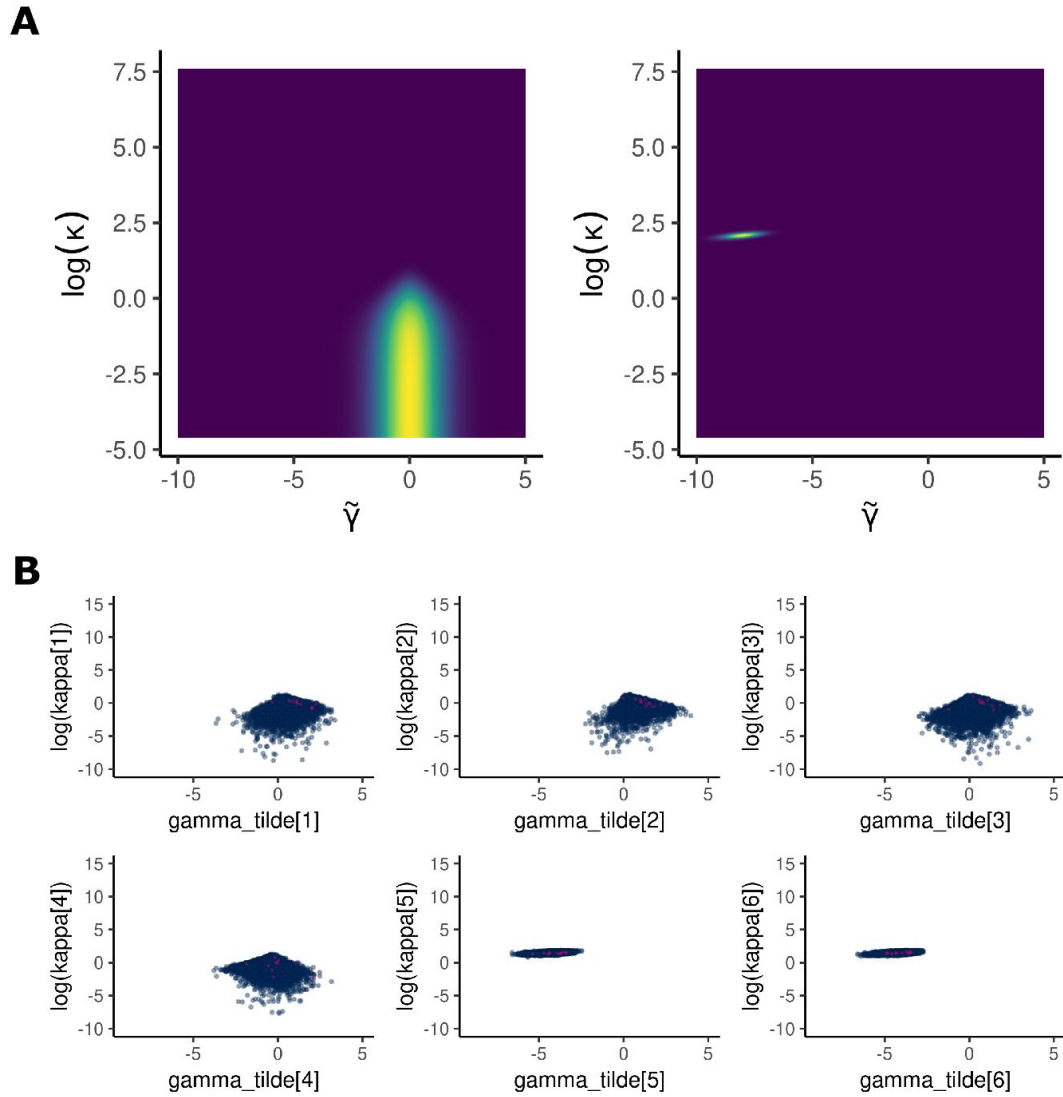


Figure 14.

Modified horseshoe densities.

A: The conditional posterior $p(\tilde{\gamma}, \kappa | \theta, \tau, y)$ when $y = 0$ (left) and $y \neq 0$ (right). **B:** MCMC pair plots of samples from the marginal posterior density $p(\tilde{\gamma}, \kappa | y)$.

where θ and τ have been fixed, is plotted when y_i lies close to the population mean (left), or when it equals zero (right). Note that the x -axis is $\tilde{\gamma}_i$ and not γ_i because the model samples a non-centered parameterisation. That is,

$$\tilde{\gamma}_i = \frac{\gamma_i - \theta}{\tau \kappa_i}, \quad \tilde{\gamma}_i \sim \text{Normal}(0, 1) \quad (37)$$

captures the deviations of γ_i around the population mean θ .

Figure 13B [↗](#) plots samples from the marginal posterior $p(\tilde{\gamma}, \kappa_i | y_i)$, when fitting to the data $\mathbf{y} = \{770, 820, 713, 541, 0, 0\}$. Each data point corresponds to one of the six plots in **Figure 13B** [↗](#). This small example dataset is in fact the cell-counts for region TMD recorded from the heterozygote group in case study 2. The similarities in geometry can be readily seen with **Figure 13A** [↗](#). A large number of divergences were produced by the sampler as demonstrated by the number of pink points compared to the non-divergent transitions in blue. For fixed τ the values $\tilde{\gamma}$ can take increase or decrease with smaller or larger κ respectively. The HalfCauchy places an extremely long right tail over κ that frustrates this relationship resulting in a posterior density that is difficult to sample from.

Modified horseshoe

For a Poisson model with parameters modeled on the log scale we consider the Cauchy parameterisation to be too extreme. In light of this we opted for a pragmatic approach, a modification to the original horseshoe by replacing the HalfCauchy distribution with a HalfNormal distribution. The modified horseshoe cuts off the top of the funnels by restricting κ to produce pleasant posterior geometry **Figure 13A, B** [↗](#). The modified horseshoe is much easier to sample from, but with the cost of a much more constraining prior over γ_i .

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Reviewer #1 (Public review):

Summary:

This work proposes a new approach to analyse cell-count data from multiple brain regions. Collecting such data can be expensive and time-intensive, so, more often than not, the dimensionality of the data is larger than the number of samples. The authors argue that Bayesian methods are much better suited to correctly analyse such data compared to classical (frequentist) statistical methods. They define a hierarchical structure, partial pooling, in which each observation contributes to the population estimate to more accurately explain the variance in the data. They present two case studies in which their method proves more

sensitive in identifying regions where there are significant differences between conditions, which otherwise would be hidden.

Strengths:

The model is presented clearly, and the advantages of the hierarchical structure are strongly justified. Two alternative ways are presented to account for the presence of zero counts. The first involves the use of a horseshoe prior, which is the more flexible option, while the second involves a modified Poisson likelihood, which is better suited to datasets with a large number of zero counts, perhaps due to experimental artifacts. The results show a clear advantage of the Bayesian method for both case studies.

The code is freely available, and it does not require a high-performance cluster to execute for smaller datasets. As Bayesian statistical methods become more accessible in various scientific fields, the whole scientific community will benefit from the transition away from p-values. Hierarchical Bayesian models are an especially useful tool that can be applied to many different experimental designs. However, while conceptually intuitive, their implementation can be difficult. The authors provide a good framework with room for improvement.

Weaknesses:

As with any Bayesian model, the choice of prior can significantly influence the results. The authors explain how the methodology can be adapted to different data properties, though selecting an appropriate prior or likelihood may not always be straightforward. They propose a 'standard workflow' as an alternative to traditional approaches, which could and should be used alongside established methods while Bayesian techniques continue to evolve and improve.

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Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

"Alternative possibilities are discussed regarding the prior and likelihood of the model. Given that the second case study inspired the introduction of the zero-inflation likelihood, it is not clear how applicable the general methodology is to various datasets. If every unique dataset requires a tailored prior or likelihood to produce the best results, the methodology will not easily replace more traditional statistical analyses that can be applied in a straightforward manner. Furthermore, the differences between the results produced by the two Bayesian models in case study 2 are not discussed. In specific regions, the models provide conflicting results (e.g., regions MH, VPMpc, RCH, SCH, etc.), which are not addressed by the authors. A third case study would have provided further evidence for the generalizability of the methodology."

We hope in this paper to propose a 'standard workflow' for these data; this standard workflow uses the horseshoe prior and we propose that this is the approach used to describe cell count data instead of the better established, but to our thinking, inefficient, t-testing approach.

The horseshoe prior is robust and allows a partially-pooled model to be used while weighing-up the contribution of different data points. This is an analogue of excluding outliers and, in any analysis it is normal to investigate further if there are points being excluded as outliers. Often this reveals a particular challenge with the data, in the case of the data here, there are a lot of

zeros, indicating that some samples should be excluded because the preparation failed to tag cells rather than because there were no cells to tag. This idea behind the ZIP example is to show that the Bayesian method can allow for this sort of further investigation and, indeed, as the reviewer notes this sort of extended analysis is often bespoke, tailored to the data.

We have clearly failed to explain that the ‘standard workflow’ we propose replace the more traditional methods is the first one we describe, with the horseshoe prior; this produces better results on both datasets than the traditional approach. However, we also feel it is useful to show how a more tailored follow-on can be useful; we need to make it clear that this is intended as an illustration of an ‘optional extra’ rather than a part of the more straightforward ‘standard workflow’.

To make this clearer we have made altered the text in several locations:

- end of Introduction: added clarifying sentence “Here, our aim is to introduce a ‘standard’ Bayesian model for cell count data. We illustrate the application of this model to two datasets, one related to neural activation and the other to developmental lineage. For the second dataset, we also demonstrate a second example extension Bayesian model.”
- Section Hierarchical modeling: “Our goal in both cases is to quantify group differences in the data. We present a ‘standard’ hierarchical model. This model reflects the experimental features common to cell count experiments and reflects the hierarchical structure of cell count data; the standard model is designed to deal robustly and efficiently with noise. On some occasions, to reflect a specific hypotheses, the structure of a particular experiment or an observed source of noise, this model can be further refined or changed to target the analysis. We will give an example of this for our second dataset.”
- Section Horseshoe prior: “The alternative is via a flexible prior such as the horseshoe Carvalho et al., 2010; Piironen and Vehtari, 2017. This more generic option may be suitable as a default ‘standard’ approach in the typical case where outliers are poorly understood.”
- Discussion: word ‘standard’ added to sentence: “Our standard workflow uses a horseshoe prior, along with the partial pooling, this allows our model to deal effectively with outliers.”
- Discussion: modified sentence “The horseshoe prior model workflow we have exhibited here is intended as a standard approach.”

Indeed, because the horseshoe prior deals robustly with outliers, whereas the ZIP is intended to model the outliers, any substantial difference between the two should be examined carefully. The referee is right to point out that we have not explained this in any detail and has helpfully listed a few brain regions where there are differences. This is useful, particularly since the examples listed illustrate in a useful way the opportunities and hazards this sort of data presents. To address this, we have added a new version of Figure 6 to the revised manuscript

Previously Figure 6 showed two example brain regions: MPN and TMd. We have now added MH and SCH to the figure, and new text commenting on the insights the plots provide, both in the Results and Discussion.

Reviewer #2 (Public review):

“A clearer link between the experimental data and model-structure terminology would be a benefit to the non-expert reader.”

This is a very good point and we are acutely aware through our own work how difficult it can be moving between fields with different research goals, different scientific cultures and different technical vocabularies. Just as it can be difficult translating from one language to

another without losing nuance and meaning, it can be a real challenge finding technical terms that are useful for the non-expert reader while retaining the precision the application requires! In the long run, we hope that, just as some of the very specialized vocabulary that surrounds frequentist statistics has become familiar to the working experimental scientists, the precise terminology involved in Bayesian modelling will become familiar and transparent. However, in advance of that day, we have included a glossary of terms at the end of the main text, and have made numerous small tweaks to make sure that link between data and model terminology is clearer and better explained.

Reviewer #1 (Recommendations from the authors):

(1) *"I would strongly recommend that the authors include more case studies in the manuscript, and address the qualitative differences between the different versions of the model."*

We agree that our method will only become established when it is applied to more datasets, we hope to contribute to further analysis and we know other people are already using the approach on their own data. We do, however, feel that adding more datasets to this paper will make it longer and more complex; the plan, instead, is to use the method on novel datasets to test specific hypotheses, so that the results will include novel scientific findings as well as adding another illustration of the Bayesian approach applied to data that is already well studied.

(2) *"Figure 6 is not discussed in the main text."*

We had discussed the results presented in Figure 6 in the second paragraph of the section "Case study two – Ontogeny of inhibitory interneurons of the mouse thalamus", however the reviewer is right in that we did not directly refer to the Figure – this was an oversight. In any case, in the revised manuscript we present a new version of Figure 6 (in response to above comment), which is now explicitly cited in the text.

Revised Figure 6: Example data and inferences highlighting model discrepancies. On the left under 'data': boxplots with medians and interquartile ranges for the raw data for four example brain regions. The shape of each point pairs left and right hemisphere readings in each of the five animals. On the right under 'inference': HDIs and confidence intervals are plotted. Purple is the Bayesian horseshoe model, pink is the Bayesian ZIP model, and orange is the sample mean. The Bayesian estimates are not strongly influenced by the zero-valued observations (MPN, SCH, TMd) or large-valued outliers (MH) and have means close to the data median. This explains the advantage of the Bayesian results over the confidence interval.

Reviewer #2 (Recommendations from the authors):

(1) *"This is a generally well-written methodology paper that also provides the underlying code as a resource. As a reviewer outside both cell-count modelling and hierarchical-Bayesian approaches (though with a general interest in the topics) I found the method a little difficult to follow and would have liked to have been left with a better understanding of how the method is applied to the data. For example, in Figure 1 we are introduced to brain region count, animal count, and "items". Then in the next line: pooling, model, structure, population and etc in subsequent lines. It is not clear what the subscripts (the pools?) are referring to: are they different regions R or animals N? These terms need to be better linked to the data and/or trimmed. Having said that, the later results look like a solid contribution to the field with a significant reduction in uncertainty from the Bayesian approach over the frequentist one. A future version of the manuscript, therefore, would benefit from greater precision of language as well as an economy and greater focus of terms linking the method to the biology. This is particularly the case*

around the exposition parts in Figure 1, Figure 2, and the “Hierarchical modelling” section.”

This is another important point. We have now made numerous small changes to tighten up the text in the paper, in response to both this point and the next point.

(2) “Language throughout could be sharpened. Subjectivity like “surprising outliers” could be removed and quirky grammar like “often small, ten is a typical” improved. There are also typos “an rate” etc that should be tidied up.”

As per previous response, we have made numerous tweaks and small improvements and feel that the paper is stronger in this respect.

(3) “Figure 1 caption. “It is a spectrum that depends” Is spectrum the right word here? Also, “thicker stroke” what does this refer to? Wasn’t immediately clear. In A, why is the whole animal within the R bracket that signifies brain regions, and then the brain regions are within the N bracket that signifies whole animals? Apart from the teal colouring, what are the other coloured regions in the image referring to? Improving this first figure would greatly help a reader unfamiliar with the context of the approach.”

We have replaced the word “spectrum” with “continuum”. We have replaced “Observed quantities have been highlighted with a thicker stroke in the graphical model.” with “The observed data quantities, y_i to y_n , are highlighted with a thick line in the model diagrams”. We have added the following text to describe the red and green lines in panel A: “green and red lines indicate regions labeled as damaged”.

(4) “On P2 there is no discussion of priors when running through the advantage of the Bayesian approach. Is this a choice or an oversight? Priors do have a role in the later analysis.”

A short additional paragraph has been added to the introduction outlining the advantage of having a prior, but also noting that the obligation to pick a prior can be intimidating and that suggesting priors is one of the contributions of our paper: “A Bayesian model also includes a set of probability distributions, referred to as the prior, which represent those beliefs it is reasonable to hold about the statistical model parameters before actually doing the experiment. The prior can be thought of as an advantage, it allows us to include in our analysis our understanding of the data based on previous experiments. The prior also makes explicit in a Bayesian model assumptions that are often implicit in other approaches. However, having to design priors is often considered a challenge and here we hope to make this more straightforward by suggesting priors that are suitable for this class of data.”

(5) “On P4 more explanation would help greatly. Formulas like $23 \times 10 \times 4$ or $50 \times 6 + 50 \times 4$ are presented without explanation. What are the various numbers being multiplied? Regions, animals? Again, a clearer link between biological data and model structure would be advantageous.”

We have now modified this line to clearly state the numbers’ sources: “The index i runs over the full set of samples, which in this case comprises 23 brain regions $\times$ 10 animals $\times$ 4 groups $\approx$ 920 datapoints in the first study, and 50 brain regions $\times$ 6 HET animals + 50 brain regions $\times$ 4 KO animals $\approx$ 500 datapoints in the second.”

(6) “P6 and Results. Is it possible to show examples of the data set sampled from? Perhaps an image or two for the two experiments. Both Figures 4 and 5 as they currently

*are could be made slightly smaller to provide space for a small explanatory sub-panel.
This would help ground the results.”*

This is a good idea. We have now added heatmap visualisations of both entire datasets to revised versions of Figures 4 and 5 (assuming that this is what the reviewer was suggesting).

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